

Time of Contact in an Elastic Collision

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INTRODUCTION

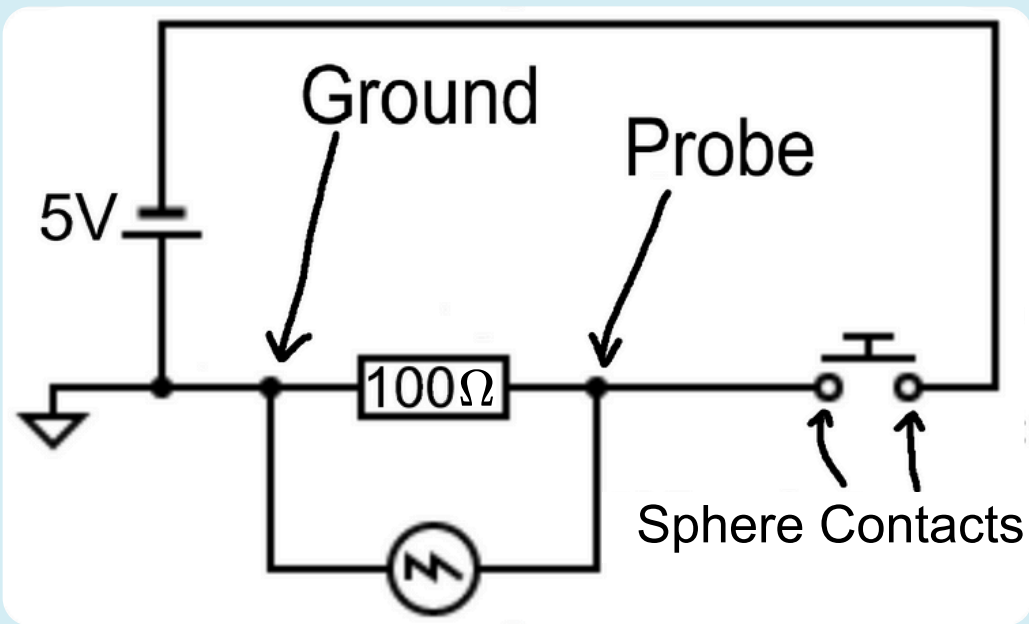
- Measure contact time during collisions of steel spheres at different impact speeds.
- Compare results with Hertzian elastic contact theory.
- Hertz predicts a power-law dependence $t = av^b$.
- For two spheres in elastic contact,

$$t = 3.78(1 - \sigma^2)^{\frac{2}{5}} \left[\frac{(R_1 + R_2)M_1^2 M_2^2}{R_1 R_2 (M_1 + M_2)^2 E^2 v} \right]^{\frac{1}{5}}$$

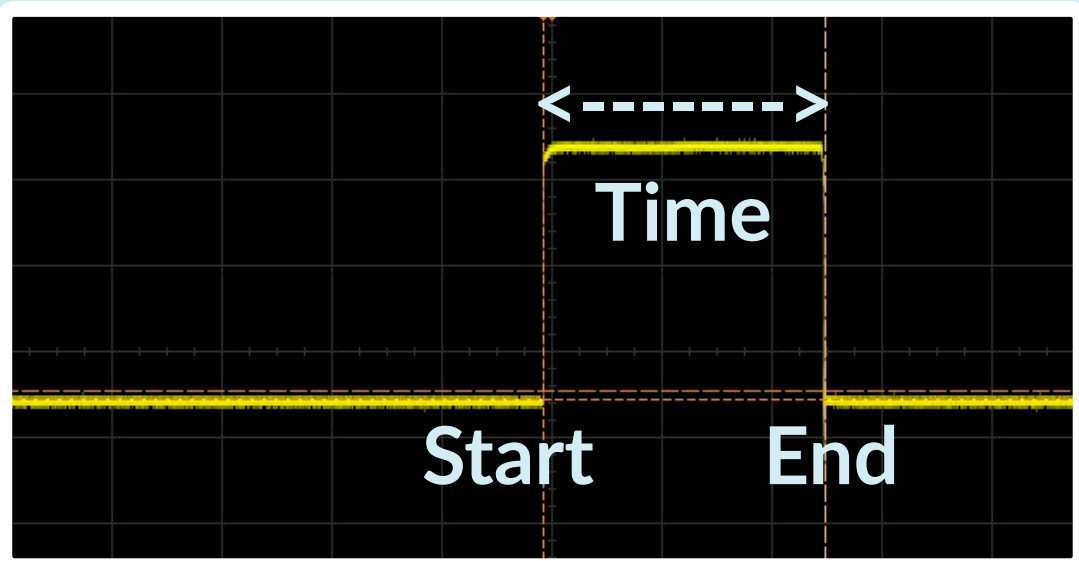
where σ is Poisson's ratio, E is Young's Modulus, and v is the relative impact speed

METHOD

- Mass and radius of spheres were measured.
- A Newton's cradle-style electrical circuit detected the start and end of contact.

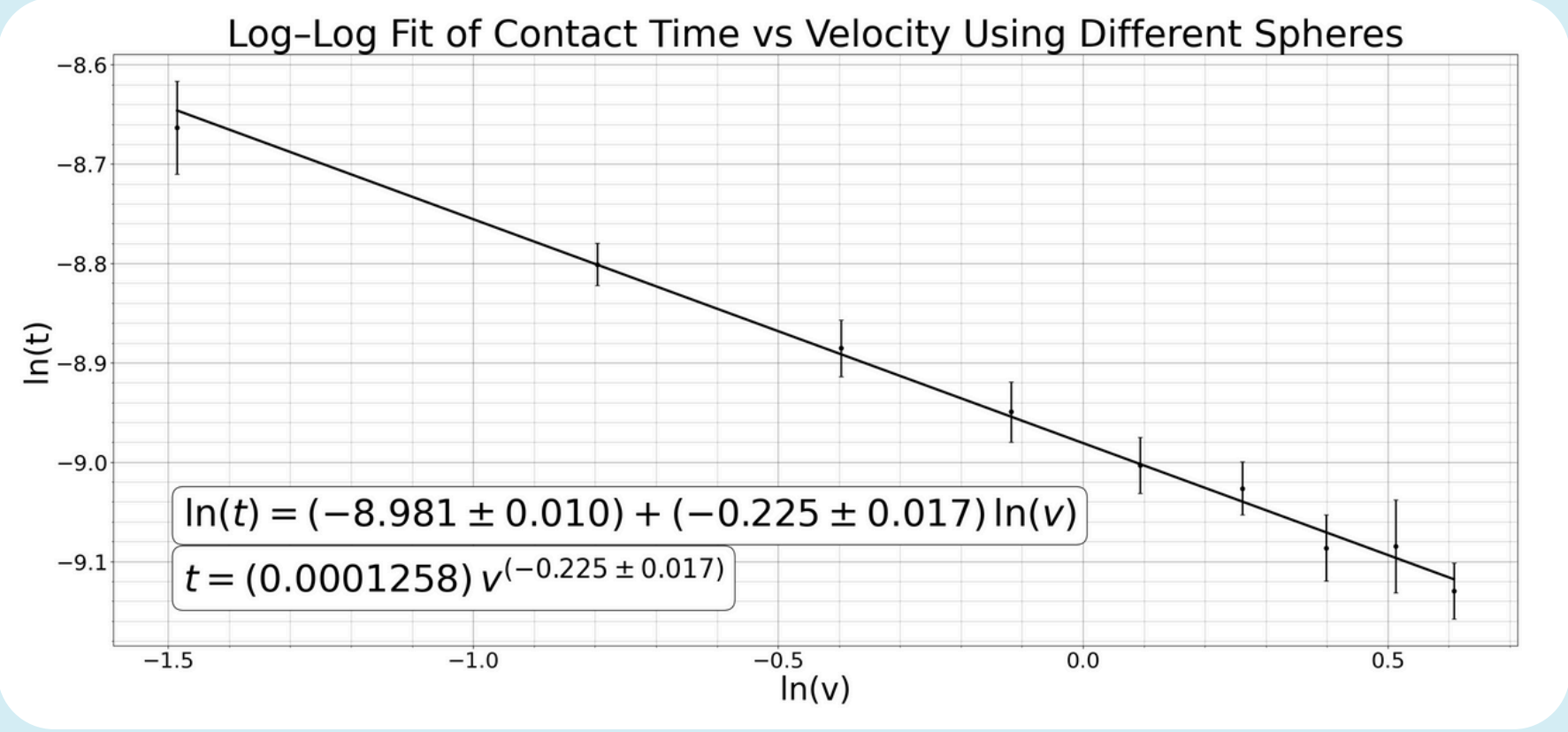
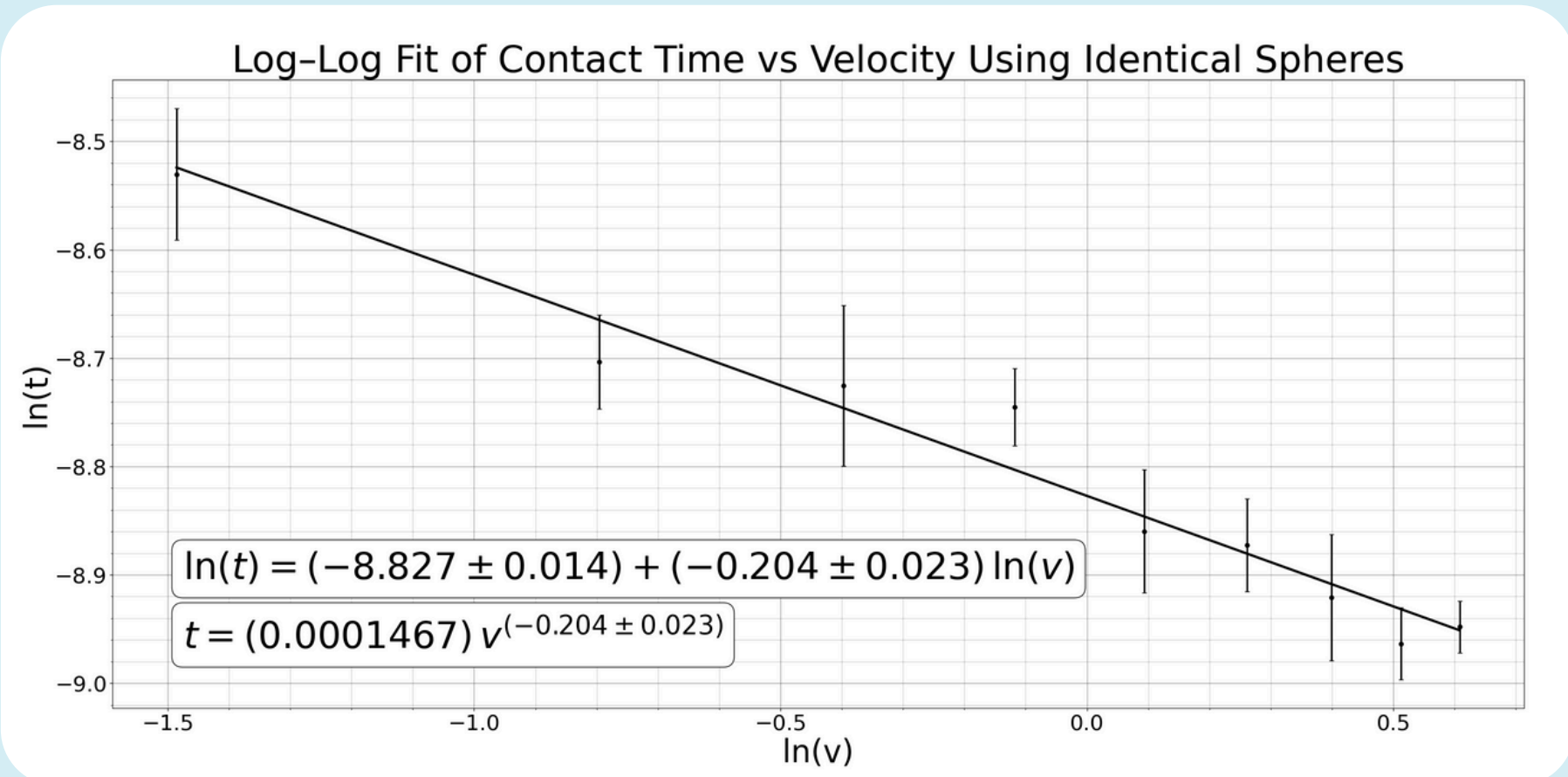


- Spheres were dropped from controlled heights to vary impact speed.
- A digital storage oscilloscope recorded the pulse width, giving the contact time for each collision.



RESULTS

Log-log fits of contact time vs velocity for identical spheres.



DISCUSSION

Identical Spheres	Different Spheres	Hertz Prediction
$a = 1.467 \times 10^{-4}$ $b = -0.204$	$a = 1.258 \times 10^{-4}$ $b = -0.225$	$a_{\text{identical}} = 1.341 \times 10^{-4}$ $a_{\text{different}} = 1.176 \times 10^{-4}$ $b = -0.200$

- Measured exponents match the Hertz prediction within uncertainty.
- Prefactors differ from theory by **9.4%** (identical spheres) and **6.6%** (different spheres), consistent with uncertainties in M , R , E , and σ .
- Remaining deviation likely from angle-reading error, release-height variation, slight misalignment of impacts, and DSO reading uncertainty.
- Overall, results show strong agreement with Hertzian elastic contact theory.

REFERENCES

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[3] Sonelastic, "Ferrous metals elastic modulus and Poisson ratio." Available at: <https://www.sonelastic.com/en/fundamentals/tables-of-materials-properties/ferrous-metals.html>